

Quotient Rings, Closed Sets, and Maximal Spectra

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Let:

- $A = \mathbb{C}[x]$, the ring of polynomials over $\mathbb{C}$ (the field of complex numbers),
- $S = \{1, x, x^2, \dots\}$, the multiplicative set generated by x , consisting of all powers of x , i.e., $S = \{x^n \mid n \geq 0\}$,
- Then, $S^{-1}A = \mathbb{C}[x, x^{-1}]$, the localization of $\mathbb{C}[x]$ at S , which gives the ring of Laurent polynomials (polynomials in x and x^{-1} with coefficients in $\mathbb{C}$).

Define a ring map:

$$f : \mathbb{C}[x] \rightarrow \mathbb{C}, \quad f(p(x)) = p(1),$$

i.e., f evaluates the polynomial at $x = 1$. Note that $f(x) = 1$, which is a unit in $\mathbb{C}$.

Claim

There exists a unique homomorphism $g : S^{-1}A \rightarrow \mathbb{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{C}[x] & \xrightarrow{f} & \mathbb{C} \\ \downarrow & & \\ S^{-1}\mathbb{C}[x] & \xrightarrow{g} & \mathbb{C} \end{array}$$

Verification

Existence of g : In $S^{-1}\mathbb{C}[x]$, every element is of the form $\frac{p(x)}{x^n}$, where $p(x) \in \mathbb{C}[x]$ and $n \geq 0$. Define:

$$g\left(\frac{p(x)}{x^n}\right) = \frac{f(p(x))}{f(x)^n}.$$

Since $f(x) = 1$, this simplifies to:

$$g\left(\frac{p(x)}{x^n}\right) = \frac{p(1)}{1^n} = p(1).$$

Commutativity of the Diagram: For any $a \in C[x]$, its image under the map $A \rightarrow S^{-1}A \rightarrow \mathbb{C}$ equals its direct image under f . For example, let $p(x) = x^2 + 3x + 2$:

$$f(p(x)) = p(1) = 1^2 + 3(1) + 2 = 6,$$

and

$$g\left(\frac{p(x)}{1}\right) = p(1) = 6.$$

Thus, the diagram commutes.

Uniqueness of g : Any other homomorphism g' must agree with f on $C[x]$ and satisfy $g'(x) = f(x) = 1$. This uniquely determines g .

Conclusion

This example illustrates the proposition in the context of $C[x]$. The ring $S^{-1}C[x]$ (the Laurent polynomials $C[x, x^{-1}]$) allows us to extend the map $f : C[x] \rightarrow \mathbb{C}$ uniquely to $g : S^{-1}C[x] \rightarrow \mathbb{C}$. The localization $S^{-1}C[x]$ captures the multiplicative inverses of elements in S , ensuring that g is well-defined and commutative.

$$\begin{array}{ccc} C[x] & & \mathbb{C} \\ f(p(x)) = p(a) & & \\ \text{localization} \downarrow \text{id} & & \\ S^{-1}C[x] = C[x]_{(x-a)} & & \mathbb{C} \end{array}$$

Here: - $f(p(x)) = p(a)$: Evaluates the polynomial at $x = a$. - $g\left(\frac{p(x)}{q(x)}\right) = \frac{p(a)}{q(a)}$ in $S^{-1}C[x] = C[x]_{(x-a)}$, focusing on the behavior near $x = a$.

2. Localization at Complement of $x - a$

$$\begin{array}{ccc} C[x] & & \mathbb{C} \\ f(p(x)) = p(a) & & \\ \text{localization} \downarrow \text{id} & & \\ S^{-1}C[x] = C[x]_{(x-a)} & & \mathbb{C} \end{array}$$

Here: - $S = C[x] \setminus (x - a)$, where S includes all polynomials not divisible by $x - a$. - $f(p(x)) = p(a)$ remains the evaluation at $x = a$. - g extends f to the localization by allowing division by elements in S .

3. Localization at Complement of Zero ($S = C[x] \setminus \{0\}$)

Proposition 1 Let $f : A \rightarrow B$ be a ring homomorphism and $S \subseteq A$ be a multiplicative set such that $f(s)$ is a unit in B for all $s \in S$. Then there is a unique homomorphism $g : S^{-1}A \rightarrow B$ such that the following diagram commutes:

